

Lab Assignment

International Cybersecurity and Digital Forensics Academy

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

1. Purpose of the Lab Assignment

The purpose of this lab is to analyze a historical wireless packet capture protected with WEP40. It focuses on identifying IEEE 802.11 traffic and protected wireless frames. The lab also demonstrates how Aircrack-ng can be used to recover or validate the WEP40 key. The capture is then decrypted offline using aircap-ng for further forensic analysis. The exercise also shows why WEP is considered insecure and should be replaced with modern wireless

2. Lab Report Structure

The report begins with the executive summary and authorization details, It covers evidence preparation, hashing, and identification of 802.11 and WEP traffic.

The report then documents key recovery or validation and offline decryption including endpoint, conversation, and recovered-object analysis.

The final sections discuss WEP security weaknesses, recommendations, and supporting evidence in the appendix.

2.1. Title Page

- Lab Title: LAB 9: WEP40 Wireless Packet Decryption and Aircrack Forensics
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- Student ID: 2025/FWSD/11320.
- Course Name: SBT-DF203: Basic Networking Skills for Digital Forensics
- Instructor Name: Aminu Idris
- Date of Submission: 25/09/2026
- Version:

2.2. Executive Summary

This lab examined a supplied historical CodeGate CTF wireless capture protected by WEP40. The evidence was preserved, decompressed, hashed, and analyzed using forensic procedures.

Aircrack-ng was used to recover or validate the known WEP40 key, followed by offline decryption with airdecap-ng. The decrypted traffic was examined for IP and MAC addresses, protocols, conversations, images, and HTML objects.

The results demonstrate the weaknesses of WEP, particularly IV reuse and RC4-related key-recovery weaknesses.

2.3. Lab Objectives

The objective is to identify management, control, and data frames within an 802.11 capture. Another objective is to understand WEP40, RC4, IV reuse, and its integrity limitations. The lab requires using Aircrack-ng and airdecap-ng on the supplied historical capture.

It also aims to extract network endpoints, protocols, images, and HTML from decrypted traffic. Finally, the lab requires explaining WEP weaknesses and recommending modern wireless security controls.

2.4. Tools and Resources Used

Item	Recommended Tool	Purpose
Operating system	Kali Linux VM	Offline capture analysis.
Wireless tools	aircrack-ng suite	Recover/validate key and decrypt capture.
Packet tools	Wireshark and tshark	Inspect encrypted/decrypted frames.
Carving tools	foremost, binwalk, file, strings	Recover transferred objects.
Compression	xz-utils	Decompress file.xz.
Evidence	Historical CodeGate CTF capture	Authorized offline dataset.

2.5. Methodology

6. Lab Folder Structure and Evidence Preparation

```
mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
```

```
cd ~/SBT-DF203-Lab9
```

```
pwd
```

```
find . -maxdepth 1 -type d -print
```

Create the folder structure before downloading or generating evidence. Store original captures under evidence and analysis copies under working.

sudo apt update

sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils

wget -O evidence/file.xz \

'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'

sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt

cp --preserve=timestamps evidence/file.xz working/file_working.xz

unxz -k working/file_working.xz

file working/file_working

sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt

```
water@kali:~$ mkdir -p ~/SBT-DF203-Lab9/{evidence,working,exported,reports,screenshots,scripts}
water@kali:~$ cd ~/SBT-DF203-Lab9
water@kali:~/SBT-DF203-Lab9$ pwd
/home/water/SBT-DF203-Lab9
water@kali:~/SBT-DF203-Lab9$ find . -maxdepth 1 -type d -print
.
./exported
./evidence
./working
./reports
./screenshots
./scripts
water@kali:~/SBT-DF203-Lab9$
```

```
water@kali:~/SBT-DF203-Lab9$ sudo apt update
[sudo] password for water:
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.6 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [54.1 MB]
Fetched 75.7 MB in 57s (1,328 kB/s)
697 packages can be upgraded. Run 'apt list --upgradable' to see them.

water@kali:~/SBT-DF203-Lab9$ sudo apt install -y aircrack-ng wireshark tshark foremost binwalk xz-utils
aircrack-ng is already the newest version (1:1.7+git20230807.4bf83fia-3).
aircrack-ng set to manually installed.
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
binwalk is already the newest version (2.4.3+dfsg1-3).
binwalk set to manually installed.
xz-utils is already the newest version (5.8.3-1).
The following packages were automatically installed and are no longer required:
amass-common libgav1-1 libkf5guiaddons-data libmbd509-7 libnsframe2 python3-kismetcapturerl433
bluez-hcidump libkf5guiaddons5 libmjpegutils-2.1-0t64 libnsisegv2 python3-kismetcapturerladsb
curlftps libgdal38 libkf5i18n-data libnng1 libnsimdf27 python3-kismetcapturerlarmr
enchant-2 libgdata-common libkf5i18n5 libnongoc-1.0-0t64 libnsimdf29 python3-multipart
firmware-marvell-proterra libgdata22 libkf5iconthemes-bin libmozjs-128-0 libnsmp40t64 python3-packaging-whl
firmware-ti-connectivity libgeoip3.13.1 libkf5iconthemes-data libmpeg2ncpp-2.1-0t64 libnsodium23 python3-pluginbase
girl1.2-girepository-2.0 libgeoipv2-2 libkf5iconthemes5 libmplex2-2.1-0t64 libnsoup-2.4-1 python3-pysmi
girl1.2-peas-1.0 libgirepository-1.0-1 libkf5itemviews-data libmupdf25.1 libnsoup2.4-common python3-pysnmp4
girl1.2-totem-1.0 libgom-1.0-0t64 libkf5itemviews5 libnet1 libspatialaudio0t64 python3-terminaltables
girl1.2-totemparser-1.0 libpgm11t64 libkf5jobwidgets-data libntfs-3g89t64 libspninxbase3t64 python3-tz
grilo-plugins-0.3 libpgmep6t64 libkf5jobwidgets5 liboath0 libswscale8 python3-wapiti-arsenic
ibus-gtk libgrilo-0.3-0 libkf5kiocore5 libobjc-14-dev libswscale8 python3-wheel-whl
kde-style-oxygen-qt6 libhdf4-0-alt libkf5kiowidgets5 libogdi4.1 libtheandctl0 python3-xlrd
kde5 libhdf5-310 libkf5kiogui5 libokularcore3 libtheora0 python3-xlutils
kf6-breeze-icon-theme libhdf5-hl-310 libkf5kiotlm5 liboxygenstyle6-6 libtheoradec1 python3-xmlt
kio libhdf5-parser2.9 libkf5kiowidgets5 liboxygenstyleconfig-data libtheoraenc1 python3-yaswp
kwayland-integration libinst-patch-1.0-2 libkf5kirigami2-5 liboxygenstyleconfig6-6 libtotoem0 python3-zombie-imp
kwayland5-data libio-pty-perl libkf5notifications-data libpackagekitqt6-1 libtsk19t64 python3.13
libabsl20240722 libipc-run-perl libkf5notifications5 libpassim1 libudfread0 python3.13-dev
libabsl20260107 libiptcdata0 libkf5service-bin libpeas-1.0-1 libunibreak6 python3.13-minimal
libann0 libjpeg-turbo-14 libkf5service-data libpeas-common libvdpau-va-gli python3.13-tk
libasmodillo4 libjs-jquery-ui libkf5services5 libperl5.40 libvldstab1.1 python3.13-venv
libaudio2 libkf5archive-data libkf5soli45 libplacebo349 libvpx0 qml-module-org-kde-kirigami2
libavfilter10 libkf5archive5 libkf5soli5-data libplacebo351 libwireshark18 qml-module-org-kde-qqc2desktopstyle
libavformat61 libkf5sonnet5-data libpocketphinx3 libwiredtap15 qml-module-org-kde-sonnet
libbabeltrace1 libkf5sauthcore5 libpopkit-qt5-1-1 libwsutil16 qml-module-qtgraphicaleffects
libblosc2-4 libkf5sonnetui5 libppopler147 libx264-164 qml-module-qtqml-models2
libbluray2 libkf5bookmarks5 libkf5style5 libportal-gtk3-1 libyelp0 qml-module-qtquick-controls2
libbluray3 libkf5codecs-data libkf5stextwidgets-data libportmidi0 libzxing3 qml-module-qtquick-layouts
libboost-chrono1.83.0t64 libkf5codecs5 libkf5stextwidgets5 libpostall1 linux-base-6.18.5-kali-amd64 qml-module-qtquick-templates2
libboost-chrono1.88.0 libkf5completion-data libkf5swallet-bin libpostproc58 linux-image-6.18.5-kali-amd64 qml-module-qtquick-window2
libboost-filesystem1.83.0 libkf5swallet-data libppsd5 document-4.0-5 llvm-spirv-18 qml-module-qtquick2
libboost-filesystem1.88.0 libkf5swallet5 libkf5swallet5 mesa-va-drivers qml6-module-org-kde-kirigamiaddons-tableview
libboost-program-options1.83.0 libkf5swaylandclient5 libpython3.13-dev mesa-vdpau-drivers qml6-module-org-kde-people
libboost-program-options1.88.0 libkf5swidgetsaddons-data libqmobipocket6-7 node-cjs-module-lexer qml6-module-qtquick-tooling
```

```
(water@kali)-[~/SBT-DF203-Lab9]
--$ wget -O evidence/file.xz 'https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz'
--2026-09-25 10:17:13-- https://raw.githubusercontent.com/ctfs/write-ups-2015/master/codegate-ctf-2015/programming/good-crypto/file.xz
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.108.133, 185.199.111.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)[185.199.109.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 12837660 (12M) [application/octet-stream]
Saving to: 'evidence/file.xz'

evidence/file.xz                               100%[=====] 12.24M  2.07MB/s  in
2026-09-25 10:17:22 (1.61 MB/s) - 'evidence/file.xz' saved [12837660/12837660]

(water@kali)-[~/SBT-DF203-Lab9]
$ ls -lh evidence/file.xz
-rw-rw-r-- 1 water water 13M Sep 25 10:17 evidence/file.xz

(water@kali)-[~/SBT-DF203-Lab9]
$
```

```
(water@kali)-[~/SBT-DF203-Lab9]
$ sha256sum evidence/file.xz | tee reports/file_xz_sha256.txt
dd54144caef34f228bf4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz

(water@kali)-[~/SBT-DF203-Lab9]
$
```

```
(water@kali)-[~/SBT-DF203-Lab9]
$ cp --preserve-timestamps evidence/file.xz working/file_working.xz

(water@kali)-[~/SBT-DF203-Lab9]
$ ls -lh evidence/file.xz working/file_working.xz
-rw-rw-r-- 1 water water 13M Sep 25 10:17 evidence/file.xz
-rw-rw-r-- 1 water water 13M Sep 25 10:17 working/file_working.xz

(water@kali)-[~/SBT-DF203-Lab9]
$ unxz -k working/file_working.xz

(water@kali)-[~/SBT-DF203-Lab9]
$ ls -lh working/
total 26M
-rw-rw-r-- 1 water water 14M Sep 25 10:17 file_working
-rw-rw-r-- 1 water water 13M Sep 25 10:17 file_working.xz

(water@kali)-[~/SBT-DF203-Lab9]
$ file working/file_working
working/file_working: pcap capture file, microsecond ts (little-endian) - version 2.4 (802.11, capture length 65535)

(water@kali)-[~/SBT-DF203-Lab9]
$ sha256sum working/file_working.xz working/file_working | tee reports/working_hashes.txt
dd54144caef34f228bf4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz
c17a3f9b95e84f5befd476dbd5c67280d1e3eea9ab402d5359cac0874ebb2d working/file_working

(water@kali)-[~/SBT-DF203-Lab9]
$
```

```
(water@kali)-[~/SBT-DF203-Lab9]
$ find . -maxdepth 2 -type f -print
./evidence/file.xz
./working/file_working
./working/file_working.xz
./reports/file_xz_sha256.txt
./reports/working_hashes.txt

(water@kali)-[~/SBT-DF203-Lab9]
$
```

```
(water@kali)-[~/SBT-DF203-Lab9]
$ ls -lh evidence working reports
evidence:
total 13M
-rw-rw-r-- 1 water water 13M Sep 25 10:17 file.xz

reports:
total 8.0K
-rw-rw-r-- 1 water water 83 Sep 25 10:19 file_xz_sha256.txt
-rw-rw-r-- 1 water water 177 Sep 25 10:32 working_hashes.txt

working:
total 26M
-rw-rw-r-- 1 water water 14M Sep 25 10:17 file_working
-rw-rw-r-- 1 water water 13M Sep 25 10:17 file_working.xz

(water@kali)-[~/SBT-DF203-Lab9]
$
```

7. Mini Evidence and Chain-of-Custody Worksheet

Field	Student Entry
Case/lab identifier	SBT-DF203-Lab9- WEP40 Wireless Packet Decryption and Aircrack Forensics
Trainee name	Olanrewaju Wale Mustapha Oladimeji
Date and time started	25/09/2026 – 5:30pm
Evidence file name(s)	evidence/file.xz; working/file_working.xz; working/file_working
Source or generation method	Supplied historical CodeGate CTF 2015 Good Crypto capture downloaded as file.xz for authorized offline forensic analysis.
Original SHA-256	dd54144caef34f228bfb4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 evidence/file.xz
Working-copy SHA-256	c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2 d working/file_working
Analysis workstation/VM	Kali Linux VM
Notes on any changes	Original file.xz was preserved in the evidence folder. A working copy was created in working, then decompressed for offline analysis. SHA-256 hashes were recorded for evidence integrity.

8. Part A - Inventory the Wireless Capture

CAP=working/file_working

capinfos "\$CAP" | tee reports/capinfos.txt

tshark -r "\$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt

tshark -r "\$CAP" -Y 'wlan' -T fields \

-e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e

```
wlan.bssid \
| head -n 100 | tee reports/wlan_frame_sample.tsv
```

```
water@kali:~/SBT-DF203-Lab9$ ls -lh working/file_working
-rw-rw-r-- 1 water water 14M Sep 25 10:17 working/file_working

water@kali:~/SBT-DF203-Lab9$ CAP=working/file_working

water@kali:~/SBT-DF203-Lab9$ echo "$CAP"
working/file_working
```

```
water@kali:~/SBT-DF203-Lab9$ capinfos "$CAP" | tee reports/capinfos.txt
File name:      working/file_working
File type:      Wireshark/tcpdump/... - pcap
File encapsulation: IEEE 802.11 Wireless LAN
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 65535 bytes
Number of packets: 45 k
File size:      14 MB
Data size:      13 MB
Capture duration: 272.969795 seconds
Earliest packet time: 2015-03-06 02:51:54.364525
Latest packet time: 2015-03-06 02:56:27.334320
Data byte rate: 49 kbps
Data bit rate: 395 kbps
Average packet size: 299.12 bytes
Average packet rate: 165 packets/s
SHA256:         c17a3f9b955e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d
SHA1:           cb929fd6e407c13639e3168b0f35f921788478d3
Strict time order: False
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = IEEE 802.11 Wireless LAN (20 - ieee-802-11)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 45169

water@kali:~/SBT-DF203-Lab9$
```

```
water@kali:~/SBT-DF203-Lab9$ tshark -r "$CAP" -q -z io,phs | tee reports/protocol_hierarchy.txt

Protocol Hierarchy Statistics
Filter:
frame                                     frames:45169 bytes:13511274
wlan                                     frames:45169 bytes:13511274
data                                     frames:15713 bytes:13104128

water@kali:~/SBT-DF203-Lab9$
water@kali:~/SBT-DF203-Lab9$ tshark -r "$CAP" -Y "wlan" -j fields -e frame.number -e frame.time -e wlan.fc.type -e wlan.fc.subtype -e wlan.sa -e wlan.da -e wlan.bssid | head -n 100 | tee reports/wlan_frame_sample.tsv
1 2015-03-06T02:51:54.364520000-0500 1 11
2 2015-03-06T02:51:54.364527000-0500 1 11
3 2015-03-06T02:51:54.364525000-0500 1 11
4 2015-03-06T02:51:54.364526000-0500 1 11
5 2015-03-06T02:51:54.365550000-0500 1 11
6 2015-03-06T02:51:54.366063000-0500 1 11
7 2015-03-06T02:51:54.366062000-0500 1 11
8 2015-03-06T02:51:54.366061000-0500 1 11
9 2015-03-06T02:51:54.366062000-0500 1 11
10 2015-03-06T02:51:54.366061000-0500 1 11
11 2015-03-06T02:51:54.366064000-0500 1 11
12 2015-03-06T02:51:54.366063000-0500 1 11
13 2015-03-06T02:51:54.366063000-0500 1 11
14 2015-03-06T02:51:54.366063000-0500 1 11
15 2015-03-06T02:51:54.366064000-0500 1 11
16 2015-03-06T02:51:54.366062000-0500 1 11
17 2015-03-06T02:51:54.366575000-0500 1 11
18 2015-03-06T02:51:54.366575000-0500 1 11
19 2015-03-06T02:51:54.366573000-0500 1 11
20 2015-03-06T02:51:54.366574000-0500 1 11
21 2015-03-06T02:51:54.366575000-0500 1 11
22 2015-03-06T02:51:54.366575000-0500 1 11
23 2015-03-06T02:51:54.366575000-0500 1 11
24 2015-03-06T02:51:54.366574000-0500 1 11
25 2015-03-06T02:51:54.366574000-0500 1 11
26 2015-03-06T02:51:54.366574000-0500 1 11
27 2015-03-06T02:51:54.366574000-0500 1 11
28 2015-03-06T02:51:54.366574000-0500 1 11
29 2015-03-06T02:51:54.366574000-0500 1 11
30 2015-03-06T02:51:54.366575000-0500 1 11
31 2015-03-06T02:51:54.366575000-0500 1 11
32 2015-03-06T02:51:54.366575000-0500 1 11
33 2015-03-06T02:51:54.366574000-0500 1 11
34 2015-03-06T02:51:54.366575000-0500 1 11
35 2015-03-06T02:51:54.366575000-0500 1 11
36 2015-03-06T02:51:54.366574000-0500 1 11
37 2015-03-06T02:51:54.366575000-0500 1 11
38 2015-03-06T02:51:54.366576000-0500 1 11
39 2015-03-06T02:51:54.366575000-0500 1 11
40 2015-03-06T02:51:54.366574000-0500 1 11
```

Frame Category	Type Value	Purpose	Example Evidence from Your Capture
Management	0	Association, authentication, beacons	Frame 54 has Type 0, Subtype 8, source MAC 00:26:66:55:97:d6, destination ff:ff:ff:ff:ff:ff, and BSSID 00:26:66:55:97:d6.
Control	1	Flow control and acknowledgements	Frames 1–53 and 55–100 contain Type 1, mainly Subtype 11, showing control traffic in the sampled frames.
Data	2	Carries network payload	The protocol hierarchy reports a data protocol entry with 15,713 frames, but the first 100-frame sample shown above does not contain Type 2.

9. Part B - Identify WEP Protection and Key Parameters

1. Confirm that protected data frames are present.

Yes protected data frames are present

```
(water@kali) ~/SBT-DF203-Lab9
$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' | wc -l
15713
(water@kali) ~/SBT-DF203-Lab9
$
```

2. Record BSSID, source/destination MAC addresses and IV-related fields where dissected.
3. Explain that 64-bit WEP normally combines a 24-bit IV with a 40-bit secret key.

64-bit WEP uses a 24-bit Initialization Vector (IV) together with a 40-bit secret key. The 24-bit IV changes between packets and is combined with the 40-bit key to create the encryption key used by WEP's RC4 encryption process. This gives a total of 64 bits (24 + 40). Because the IV is very short, it can eventually be reused, allowing attackers to collect packets with repeated IVs and exploit weaknesses in WEP. Therefore, IV reuse is one of the major weaknesses of WEP40.

4. Explain why repeated IVs and RC4 weaknesses make WEP unsuitable.

Repeated IVs make WEP weak because the IV is only 24 bits, so the same IV can eventually be reused in different packets. When this happens, attackers can compare encrypted packets and learn information about the RC4 keystream. WEP also uses RC4 in a weak way, making it possible to exploit patterns and recover the secret key from enough captured traffic. This means that collecting wireless packets can provide enough information to break the WEP protection. Therefore, WEP is unsuitable for protecting modern wireless networks and should be replaced with WPA2-AES/CCMP or WPA3.


```
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields \
  -e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e
wlan.wep.key \
  | head -n 200 | tee reports/wep_protected_frames.tsv
```

```
tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv \
  | sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
```

```
(water@kali) [~/SBT-DF203-Lab9]
$ ls -lh working/file_working
-rw-rw-r-- 1 water water 14M Sep 25 10:17 working/file_working

(water@kali) [~/SBT-DF203-Lab9]
$ CAP=working/file_working

(water@kali) [~/SBT-DF203-Lab9]
$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e frame.number -e frame.time_epoch -e wlan.sa -e wlan.da -e wlan.bssid -e wlan.wep.iv -e wlan.wep.key | head -n 200 | tee reports/wep_protected_f
rames.tsv
9432  1425628319.232982000  00:26:66:55:97:d4  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0x67ffec  0
10130 1425628320.138252000  f0:f6:1c:68:96:7c  00:26:66:55:97:d6  00:26:66:55:97:d6  0xc2cbeb  0
10155 1425628320.176656000  f0:f6:1c:68:96:7c  04:8d:38:48:e5:b4  00:26:66:55:97:d6  0xc3cbeb  0
10157 1425628320.177168000  f0:f6:1c:68:96:7c  04:8d:38:48:a8:b5  00:26:66:55:97:d6  0xc4cbeb  0
10159 1425628320.177168000  f0:f6:1c:68:96:7c  08:10:77:92:7c:2f  00:26:66:55:97:d6  0xc5cbeb  0
10161 1425628320.177168000  f0:f6:1c:68:96:7c  00:26:66:55:97:d4  00:26:66:55:97:d6  0xc6cbeb  0
10163 1425628320.177168000  f0:f6:1c:68:96:7c  c8:3a:35:58:c4:ba  00:26:66:55:97:d6  0xc7cbeb  0
10165 1425628320.177167000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xc8cbeb  0
10167 1425628320.177686000  00:26:66:55:97:d4  f0:f6:1c:68:96:7c  00:26:66:55:97:d6  0xc9cbeb  0
10170 1425628320.180760000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd0cbeb  0
10171 1425628320.180976000  f0:f6:1c:68:96:7c  33:33:00:00:00:02  00:26:66:55:97:d6  0xd1cbeb  0
10172 1425628320.187417000  00:26:66:55:97:d4  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd2cbeb  0
10173 1425628320.187408000  f0:f6:1c:68:96:7c  33:33:00:00:00:02  00:26:66:55:97:d6  0xd3cbeb  0
10175 1425628320.187919000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd4cbeb  0
10177 1425628320.188952000  f0:f6:1c:68:96:7c  33:33:00:00:00:02  00:26:66:55:97:d6  0xd5cbeb  0
10178 1425628320.189976000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd6cbeb  0
10183 1425628320.192016000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd7cbeb  0
10185 1425628320.193048000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd8cbeb  0
10224 1425628320.259088000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd9cbeb  0
10228 1425628320.262680000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xdacbeb  0
10229 1425628320.262678000  00:26:66:55:97:d4  f0:f6:1c:68:96:7c  00:26:66:55:97:d6  0xdbcbeb  0
10233 1425628320.263184000  f0:f6:1c:68:96:7c  00:26:66:55:97:d4  00:26:66:55:97:d6  0xdcbeb  0
10261 1425628320.419350000  f0:f6:1c:68:96:7c  00:26:66:55:97:d4  00:26:66:55:97:d6  0xdccbeb  0
10266 1425628320.420374000  f0:f6:1c:68:96:7c  01:00:5e:00:00:02  00:26:66:55:97:d6  0xdccbeb  0
10268 1425628320.420374000  f0:f6:1c:68:96:7c  01:00:5e:00:00:fb  00:26:66:55:97:d6  0xdccbeb  0
10270 1425628320.420866000  f0:f6:1c:68:96:7c  01:00:5e:00:00:02  00:26:66:55:97:d6  0xdccbeb  0
10272 1425628320.422421000  f0:f6:1c:68:96:7c  01:00:5e:00:00:fb  00:26:66:55:97:d6  0xdccbeb  0
10353 1425628320.583190000  00:26:66:55:97:d4  f0:f6:1c:68:96:7c  00:26:66:55:97:d6  0xdccbeb  0
10355 1425628320.583200000  00:26:66:55:97:d4  f0:f6:1c:68:96:7c  00:26:66:55:97:d6  0xdccbeb  0
```

```
(water@kali) [~/SBT-DF203-Lab9]
$ head -n 20 reports/wep_protected_frames.tsv
9432  1425628319.232982000  00:26:66:55:97:d4  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0x67ffec  0
10130 1425628320.138252000  f0:f6:1c:68:96:7c  00:26:66:55:97:d6  00:26:66:55:97:d6  0xc2cbeb  0
10155 1425628320.176656000  f0:f6:1c:68:96:7c  04:8d:38:48:e5:b4  00:26:66:55:97:d6  0xc3cbeb  0
10157 1425628320.177168000  f0:f6:1c:68:96:7c  04:8d:38:48:a8:b5  00:26:66:55:97:d6  0xc4cbeb  0
10159 1425628320.177168000  f0:f6:1c:68:96:7c  08:10:77:92:7c:2f  00:26:66:55:97:d6  0xc5cbeb  0
10161 1425628320.177168000  f0:f6:1c:68:96:7c  00:26:66:55:97:d4  00:26:66:55:97:d6  0xc6cbeb  0
10163 1425628320.177168000  f0:f6:1c:68:96:7c  c8:3a:35:58:c4:ba  00:26:66:55:97:d6  0xc7cbeb  0
10165 1425628320.177167000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xc8cbeb  0
10167 1425628320.177686000  00:26:66:55:97:d4  f0:f6:1c:68:96:7c  00:26:66:55:97:d6  0xc9cbeb  0
10170 1425628320.180760000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd0cbeb  0
10171 1425628320.182780000  f0:f6:1c:68:96:7c  33:33:00:00:00:02  00:26:66:55:97:d6  0xd1cbeb  0
10172 1425628320.187417000  00:26:66:55:97:d4  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd2cbeb  0
10173 1425628320.187408000  f0:f6:1c:68:96:7c  33:33:00:00:00:02  00:26:66:55:97:d6  0xd3cbeb  0
10175 1425628320.187919000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd4cbeb  0
10177 1425628320.188952000  f0:f6:1c:68:96:7c  33:33:00:00:00:02  00:26:66:55:97:d6  0xd5cbeb  0
10178 1425628320.189976000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd6cbeb  0
10183 1425628320.192016000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd7cbeb  0
10185 1425628320.193048000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd8cbeb  0
10224 1425628320.259088000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xd9cbeb  0
10228 1425628320.262680000  f0:f6:1c:68:96:7c  ff:ff:ff:ff:ff:ff  00:26:66:55:97:d6  0xdacbeb  0

(water@kali) [~/SBT-DF203-Lab9]
$ tshark -r "$CAP" -Y 'wlan.fc.protected==1' -T fields -e wlan.wep.iv | sort | uniq -c | sort -nr | head -n 30 | tee reports/repeated_iv_summary.txt
 7 0x42e8eb
 7 0x28d3eb
 6 0x95dfeb
 5 0x9415ed
 5 0x86e5eb
 5 0x7ae5eb
 5 0x28e7eb
 5 0x19e5eb
 4 0xd5d7eb
 4 0xd3e8eb
 4 0xc915ed
 4 0xc7d8eb
 4 0xad44eb
 4 0x73e5eb
 4 0x65d8eb
 4 0x4ad5eb
 4 0x31e7eb
 4 0x2723ed
 4 0x10e7eb
 3 0xffe5eb
 3 0xf609ed
 3 0xa307ed
 3 0x97ddeb
 3 0x96d1eb
 3 0x92e8eb
 3 0x91e8eb
 3 0x8adeeb
 3 0x630aed
```


10. Part C - Recover or Validate the WEP40 Key

Run Aircrack-ng against the supplied capture. Record selected network/BSSID, number of IVs, key length and recovered key. The slide-provided key is A4:3D:F6:F3:74, entered without colons for airdecap-ng.

```
aircrack-ng "$CAP" | tee reports/aircrack_output.txt
```

```
# Expected historical training key from the slide: A4:3D:F6:F3:74
```

```
printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
```

```
(water@kali)-[~/SBT-DF203-Lab9]
$ ls -lh "$CAP"
-rw-rw-r-- 1 water water 14M Sep 25 10:17 working/file_working
Got 15477 out of 15000 IVsStarting PTW attack with 15477 ivs.

(water@kali)-[~/SBT-DF203-Lab9]
$ aircrack-ng "$CAP" | tee reports/aircrack_output.txt
Reading packets, please wait...
Opening working/file_working
Read 45169 packets.

# BSSID      ESSID      Encryption
1 00:26:66:55:97:D6  cgnetwork  WEP (15477 IVs)

Choosing first network as target.

Reading packets, please wait...
Opening working/file_working
Read 45169 packets.

1 potential targets
Attack will be restarted every 5000 captured ivs.
```

```
Aircrack-ng 1.7

[00:00:00] Tested 83 keys (got 15477 IVs)

KB  depth  byte(vote)
0  0/ 1  A4(22784) 62(20992) A8(19968) 86(19968) 42(19456) 6E(19456) 91(19200) 87(19200) 26(18944) 68(18944) 9E(18944) 0E(18688) 74(18688) E4(18688) 0A(18432) 00(18432)
1  0/ 1  3D(23044) 51(20736) 07(20480) 62(19968) 78(19968) 1F(19712) 80(19712) 8D(19456) 80(19200) 85(19200) 9D(19200) 13(18944) EC(18944) 00(18688) 98(18688) 24(18432)
2  0/ 1  F6(23808) E4(20992) D0(20736) 68(20224) 95(19712) 38(19456) 0C(19200) 45(18944) 4F(18944) A8(18944) F7(18944) 60(18688) 63(18688) A8(18688) AD(18688) 8B(18688)
3  1/ 10 F3(20480) C5(19968) D0(19968) 3E(19712) 43(19456) 52(19456) B2(19456) 09(19456) 20(19456) 8F(19200) A0(19200) B0(19200) 04(18944) 2C(18944) 8D(18944) 43(18688)
4  6/ 9  01(19712) 20(19456) 3E(19456) 52(19456) 5C(19456) 90(19456) F9(19456) 45(18944) 85(18944) 95(18944) D3(18944) 3A(18688) 40(18688) 9D(18688) C8(18688) 1E(18432)

KEY FOUND! [ A4:3D:F6:F3:74 ]
Decrypted correctly: 100%

(water@kali)-[~/SBT-DF203-Lab9]
```

```
(water@kali)-[~/SBT-DF203-Lab9]
$ printf 'A4:3D:F6:F3:74\n' | tee reports/validated_wep40_key_masked.txt
A4:3D:F6:F3:74

(water@kali)-[~/SBT-DF203-Lab9]
```

11. Part D - Decrypt the Capture Offline

```
cd working
```

```
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
```

```
cd ..
```

```
find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
```

```
sha256sum working/* | tee reports/all_working_file_hashes.txt
```

Identify the file created by airdecap-ng, usually with a -dec suffix. Open it in Wireshark and confirm that IP/TCP/HTTP or other higher-layer protocols are now visible.

```
(water@kali)-[~/SBT-DF203-Lab9]
└─$ ls -lh working/file_working
-rw-rw-r-- 1 water water 14M Sep 25 10:17 working/file_working

(water@kali)-[~/SBT-DF203-Lab9]
└─$ cd working
airdecap-ng -w A43DF6F374 file_working | tee ../reports/airdecap_output.txt
Total number of stations seen      10
Total number of packets read      45169
Total number of WEP data packets  15477
Total number of WPA data packets   0
Number of plaintext data packets   0
Number of decrypted WEP packets    15477
Number of corrupted WEP packets    0
Number of decrypted WPA packets    0
Number of bad TKIP (WPA) packets   0
Number of bad CCMP (WPA) packets   0

(water@kali)-[~/SBT-DF203-Lab9/working]
└─$
```

```
(water@kali)-[~/SBT-DF203-Lab9/working]
└─$ cd ..

(water@kali)-[~/SBT-DF203-Lab9]
└─$ find working -maxdepth 1 -type f -ls | tee reports/decrypted_files_inventory.txt
1610655 13904 -rw-rw-r-- 1 water water 14234002 Sep 25 10:17 working/file_working
1610622 12540 -rw-rw-r-- 1 water water 12837660 Sep 25 10:17 working/file_working.xz
1610735 12488 -rw-rw-r-- 1 water water 12784501 Sep 25 16:50 working/file_working-dec

(water@kali)-[~/SBT-DF203-Lab9]
└─$ sha256sum working/* | tee reports/all_working_file_hashes.txt
c17a3f9b95e84f5befd476dbd55c67286d1e3eea9ab402d5359cac0874ebb2d working/file_working
167c91994c26977f9048227deb9882caf3cf3c763977f2059604f9a6a40b04 working/file_working-dec
dd54144caef34f228bf4b87a9101ca7173969376f7ed357bd068061d4f4b8d6 working/file_working.xz

(water@kali)-[~/SBT-DF203-Lab9]
└─$
```

```
(water@kali)-[~/SBT-DF203-Lab9]
└─$ tshark -r "SDEC" -q -z io,phs | tee reports/decrypted_protocol_hierarchy.txt

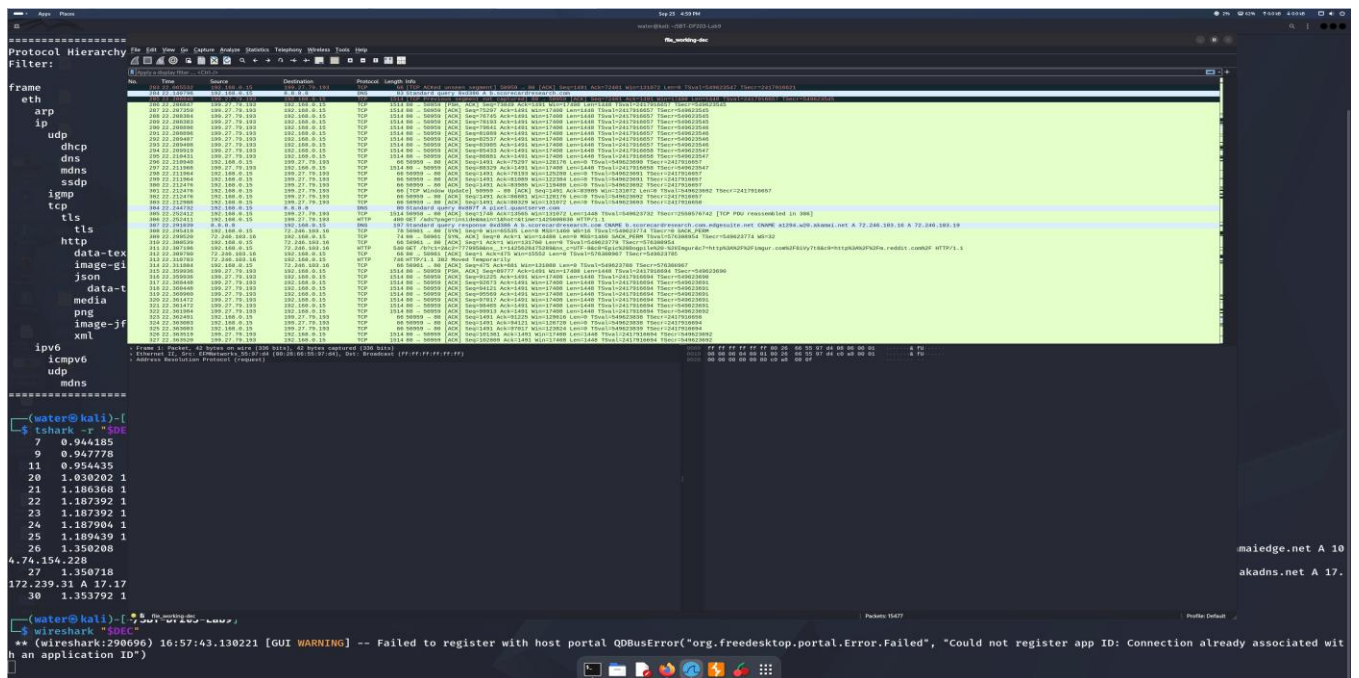
=====
Protocol Hierarchy Statistics
Filter:

frame
eth
arp
ip
udp
dhcp
dns
mdns
ssdp
igmp
tcp
tls
http
data-text-lines
image-gif
json
data-text-lines
media
png
image-jfif
xml
ipv6
icmpv6
udp
mdns
frames:15477 bytes:12536845
frames:15477 bytes:12536845
frames:60 bytes:2610
frames:15361 bytes:12525585
frames:159 bytes:311074
frames:7 bytes:2890
frames:81 bytes:10351
frames:40 bytes:6556
frames:31 bytes:11277
frames:7 bytes:364
frames:15195 bytes:12494147
frames:138 bytes:82995
frames:13 bytes:11087
frames:230 bytes:134537
frames:28 bytes:23147
frames:7 bytes:3461
frames:15 bytes:3982
frames:1 bytes:813
frames:22 bytes:17622
frames:10 bytes:5761
frames:1 bytes:435
frames:1 bytes:313
frames:50 bytes:8650
frames:20 bytes:1628
frames:36 bytes:7022
frames:36 bytes:7022

=====

(water@kali)-[~/SBT-DF203-Lab9]
└─$ tshark -r "SDEC" -Y 'ip || tcp || http' -c 30
  7  0.944185      0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e6be1c
  9  0.947778      0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e6be1c
11  0.954435     192.168.0.1 → 255.255.255.255 DHCP 590 DHCP NAK - Transaction ID 0x21e6be1c
20  1.030202     192.168.0.15 → 8.8.8.8      DNS 73 Standard query 0x9583 A www.apple.com
21  1.186368     192.168.0.15 → 8.8.8.8      DNS 85 Standard query 0xb40 A 48-courier.push.apple.com
22  1.187392     192.168.0.15 → 224.0.0.2    IGMPv2 46 Leave Group 224.0.0.251
23  1.187392     192.168.0.15 → 224.0.0.251    IGMPv2 46 Membership Report group 224.0.0.251
24  1.187904     192.168.0.15 → 224.0.0.2    IGMPv2 46 Leave Group 224.0.0.251
25  1.189439     192.168.0.15 → 224.0.0.251    IGMPv2 46 Membership Report group 224.0.0.251
26  1.350208      8.8.8.8 → 192.168.0.15 DNS 203 Standard query response 0x9583 A www.apple.com CNAME www.isg-apple.com.akadns.net CNAME www.apple.com.edgekey.net CNAME e3191.dscc.akamaiedge.net A 10.4.74.154.228
27  1.350718      8.8.8.8 → 192.168.0.15 DNS 299 Standard query response 0xb40 A 48-courier.push.apple.com CNAME 48-courier.push-apple.com.akadns.net CNAME us-courier.push-apple.com.akadns.net A 17.172.239.31 A 17.172.232.188 A 17.172.239.43 A 17.172.232.166 A 17.172.239.25 A 17.172.232.170 A 17.172.232.185
30  1.353792     192.168.0.15 → 17.172.239.31 TCP 70 50954 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=16 TSval=549602880 TSecr=0 SACK_PERM

(water@kali)-[~/SBT-DF203-Lab9]
└─$
```



12. Part E - Extract Endpoints and Conversations

DEC=\$(find working -maxdepth 1 -type f -name '*-dec*' | head -n 1)

echo "Decrypted capture: \$DEC" | tee reports/decrypted_capture_path.txt

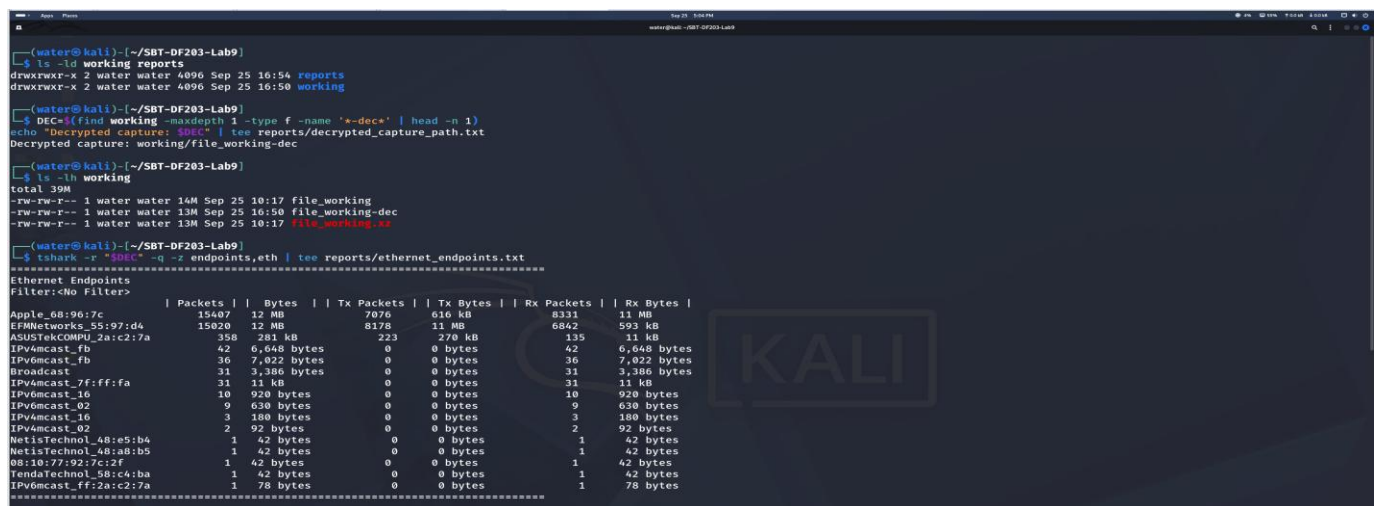
tshark -r "\$DEC" -q -z endpoints,eth | tee reports/ethernet_endpoints.txt

tshark -r "\$DEC" -q -z endpoints,ip | tee reports/ip_endpoints.txt

tshark -r "\$DEC" -q -z conv,tcp | tee reports/tcp_conversations.txt

tshark -r "\$DEC" -Y 'ip' -T fields -e frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol \

| head -n 200 | tee reports/ip_mac_mapping_sample.tsv



```

(water@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "506c" -q -z endpoints,ip | tee reports/ip_endpoints.txt
=====
IPv4 Endpoints
Filter: <No Filter>

| Packets | | Bytes | | Tx Packets | | Tx Bytes | | Rx Packets | | Rx Bytes |
192.168.0.15 15309 12 MB 6995 608 kB 8314 11 MB
198.90.20.111 8057 6,916 kB 4446 6,637 kB 3611 279 kB
199.27.79.193 5094 4,825 kB 3131 4,003 kB 2563 221 kB
192.168.0.9 329 277 kB 197 265 kB 132 11 kB
173.194.127.141 153 109 kB 80 102 kB 73 7,364 bytes
173.194.127.55 143 89 kB 71 82 kB 72 6,990 bytes
8.8.8.8 81 10 kB 40 6,960 bytes 41 3,391 bytes
199.16.158.186 66 15 kB 29 10 kB 37 5,214 bytes
173.194.127.227 66 24 kB 26 15 kB 40 8,455 bytes
173.194.127.229 55 22 kB 23 13 kB 32 9,731 bytes
31.13.70.1 51 19 kB 26 15 kB 25 3,151 bytes
173.241.248.220 44 6,848 bytes 20 3,122 bytes 24 3,726 bytes
198.41.208.140 44 28 kB 21 25 kB 23 2,718 bytes
224.0.0.251 42 6,648 bytes 0 0 bytes 42 6,648 bytes
173.194.127.237 36 9,526 bytes 14 6,435 bytes 22 3,091 bytes
216.58.220.234 34 15 kB 16 12 kB 18 2,497 bytes
192.168.0.1 33 12 kB 33 12 kB 0 0 bytes
173.241.248.219 33 5,733 bytes 14 2,807 bytes 19 2,926 bytes
104.74.158.12 33 21 kB 18 20 kB 15 1,236 bytes
74.125.203.156 32 20 kB 16 17 kB 16 2,534 bytes
239.255.255.250 31 11 kB 0 0 bytes 31 11 kB
173.194.127.231 31 6,816 bytes 12 5,197 bytes 19 1,619 bytes
198.41.209.137 28 16 kB 13 14 kB 8 1,556 bytes
17.172.239.31 25 6,504 bytes 12 3,973 bytes 13 2,531 bytes
173.194.72.95 23 4,077 bytes 9 2,232 bytes 14 1,845 bytes
173.194.127.122 21 12 kB 11 11 kB 10 1,759 bytes
69.59.169.185 18 5,604 bytes 7 3,675 bytes 11 1,069 bytes
72.246.103.16 16 3,697 bytes 6 1,732 bytes 10 1,965 bytes
23.23.127.192 15 3,383 bytes 6 1,370 bytes 9 2,013 bytes
173.194.127.198 14 4,480 bytes 6 2,931 bytes 8 1,549 bytes
173.194.122.24 14 1,724 bytes 5 688 bytes 9 1,036 bytes
23.21.133.99 13 2,314 bytes 5 880 bytes 8 1,424 bytes
50.31.251.34 12 3,483 bytes 6 2,777 bytes 6 706 bytes
108.168.254.139 11 2,655 bytes 5 1,690 bytes 6 965 bytes
108.168.254.138 11 2,333 bytes 5 1,124 bytes 6 1,209 bytes
54.240.248.207 10 1,217 bytes 4 441 bytes 6 776 bytes
173.194.127.100 10 1,921 bytes 4 715 bytes 6 1,206 bytes
54.169.42.244 10 2,054 bytes 4 1,054 bytes 6 1,000 bytes
173.194.127.119 10 1,534 bytes 4 670 bytes 6 864 bytes
173.194.127.128 9 1,468 bytes 3 604 bytes 6 864 bytes
23.21.196.184 8 975 bytes 2 140 bytes 6 835 bytes
203.190.124.19 8 1,742 bytes 3 648 bytes 5 1,094 bytes
255.255.255.255 6 2,300 bytes 0 0 bytes 6 2,300 bytes
0.0.0.0 5 1,710 bytes 5 1,710 bytes 0 0 bytes
173.194.127.173 4 240 bytes 2 132 bytes 2 108 bytes
17.172.239.127 3 206 bytes 1 74 bytes 2 132 bytes
173.194.122.15 3 206 bytes 1 74 bytes 2 132 bytes

```

```

(water@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "506c" -q -z conv,tcp | tee reports/tcp_conversations.txt
=====
TCP Conversations
Filter: <No Filter>

| Frames | Bytes | | Frames | Bytes | | Total | Relative | Duration |
| Frames | Bytes | | Frames | Bytes | | Frames | Bytes | Start | End |
192.168.0.15:51007 <--> 198.90.20.111:80 1257 1,885 kB 978 73 kB 2235 1,958 kB 113.588792000 45.4035
192.168.0.15:50959 <--> 199.27.79.193:80 1209 1,816 kB 959 71 kB 2168 1,887 kB 21.074236000 34.8633
192.168.0.15:51008 <--> 198.90.20.111:80 1159 1,739 kB 947 70 kB 2106 1,810 kB 113.588792000 46.0179
192.168.0.15:51001 <--> 199.27.79.193:80 1145 1,725 kB 927 66 kB 2072 1,791 kB 63.060042000 26.6888
192.168.0.15:51009 <--> 198.90.20.111:80 687 1,028 kB 572 44 kB 1259 1,072 kB 113.605175000 20.7933
192.168.0.15:51014 <--> 198.90.20.111:80 664 991 kB 566 44 kB 1230 1,036 kB 114.020474000 25.0538
192.168.0.15:51006 <--> 198.90.20.111:80 654 970 kB 520 40 kB 1174 1,011 kB 113.578039000 25.4942
192.168.0.15:51000 <--> 199.27.79.193:80 375 561 kB 323 25 kB 698 586 kB 58.731642000 4.4552
192.168.0.15:50999 <--> 199.27.79.193:80 208 303 kB 180 15 kB 386 319 kB 56.792636000 1.9390
192.168.0.15:51036 <--> 192.168.0.9:5080 98 144 kB 70 5,705 bytes 168 150 kB 243.325613000 0.1777
192.168.0.15:50958 <--> 199.27.79.193:80 73 93 kB 67 12 kB 140 105 kB 20.145401000 14.9846
192.168.0.15:51035 <--> 192.168.0.9:5080 80 116 kB 56 4,756 bytes 136 121 kB 243.281602000 0.1398
192.168.0.15:50983 <--> 173.194.127.141:80 71 99 kB 59 4,392 bytes 130 104 kB 27.831547000 31.4188
192.168.0.15:50997 <--> 199.27.79.193:80 62 87 kB 37 3,826 bytes 99 90 kB 56.126972000 11.0492
192.168.0.15:51034 <--> 173.194.127.229:443 23 13 kB 32 9,731 bytes 55 22 kB 234.384001000 6.7568
192.168.0.15:51005 <--> 198.90.20.111:80 25 22 kB 28 4,757 bytes 53 27 kB 113.019956000 2.8111
192.168.0.15:50960 <--> 31.13.70.1:443 26 15 kB 25 3,151 bytes 51 19 kB 21.329724000 32.8418
192.168.0.15:51028 <--> 173.194.127.227:443 18 11 kB 19 7,603 bytes 47 18 kB 172.217100000 37.9452
192.168.0.15:51003 <--> 198.41.208.140:80 21 25 kB 23 2,710 bytes 44 28 kB 102.780857000 32.8078
192.168.0.15:51018 <--> 173.194.127.55:80 4 20 24 kB 16 1,494 bytes 36 26 kB 115.102391000 30.4286
192.168.0.15:50973 <--> 199.16.158.186:443 15 5,156 bytes 19 2,640 bytes 34 7,796 bytes 25.665085000 31.5233
192.168.0.15:51002 <--> 104.74.158.12:80 18 20 kB 15 1,236 bytes 33 21 kB 63.175099000 30.4746
192.168.0.15:50974 <--> 199.16.158.186:443 14 5,082 bytes 18 2,574 bytes 32 7,656 bytes 25.665085000 31.5223
192.168.0.15:51016 <--> 173.194.127.55:80 13 15 kB 19 1,740 bytes 32 16 kB 115.101879000 30.4439
192.168.0.15:51020 <--> 74.125.203.156:80 16 17 kB 16 2,534 bytes 32 20 kB 115.146935000 31.5115
192.168.0.15:50956 <--> 198.41.209.137:80 13 14 kB 15 1,556 bytes 28 16 kB 13.669693000 31.2704
192.168.0.15:51032 <--> 173.194.127.231:443 11 5,123 bytes 16 1,409 bytes 27 6,532 bytes 210.917057000 0.2191
192.168.0.15:50954 <--> 17.172.239.31:443 13 2,531 bytes 13 2,531 bytes 25 6,504 bytes 11.353792000 2.6631
192.168.0.15:51015 <--> 173.194.127.55:80 13 14 kB 12 1,230 bytes 25 15 kB 115.100855000 30.4291
192.168.0.15:51017 <--> 173.194.127.55:80 13 14 kB 12 1,221 bytes 25 15 kB 115.101880000 30.4306
192.168.0.15:51019 <--> 173.194.127.55:80 12 14 kB 13 1,296 bytes 25 15 kB 115.102904000 30.4470
192.168.0.15:50989 <--> 173.194.127.237:443 9 4,925 bytes 15 2,069 bytes 24 6,994 bytes 28.751611000 30.5064
192.168.0.15:50980 <--> 173.194.127.122:80 11 11 kB 10 1,759 bytes 21 12 kB 27.325115000 30.8808
192.168.0.15:51010 <--> 216.58.220.234:80 9 7,719 bytes 10 1,528 bytes 19 9,247 bytes 113.678904000 32.9831
192.168.0.15:51026 <--> 173.194.127.227:443 8 4,850 bytes 11 852 bytes 19 5,702 bytes 171.717324000 0.6262
192.168.0.15:50967 <--> 199.27.79.193:80 8 1,592 bytes 10 5,118 bytes 18 6,710 bytes 25.133117000 11.0763
192.168.0.15:50995 <--> 199.27.79.193:80 9 4,667 bytes 9 3,658 bytes 18 8,325 bytes 55.735230000 11.2719
192.168.0.15:51013 <--> 69.59.169.185:80 7 3,675 bytes 11 1,969 bytes 18 5,644 bytes 113.701432000 31.9258
192.168.0.15:50961 <--> 72.246.103.16:80 6 1,732 bytes 10 1,965 bytes 16 3,697 bytes 22.295419000 56.6984
192.168.0.15:50969 <--> 199.27.79.193:80 7 1,006 bytes 9 2,952 bytes 16 3,958 bytes 25.167934000 12.1459
192.168.0.15:50966 <--> 23.23.127.192:80 6 1,370 bytes 9 2,013 bytes 15 3,383 bytes 24.785469000 31.3185
192.168.0.15:51021 <--> 216.58.220.234:80 7 5,284 bytes 8 969 bytes 15 6,223 bytes 115.526841000 31.1302
192.168.0.15:50964 <--> 199.27.79.193:80 7 3,129 bytes 7 2,304 bytes 14 5,433 bytes 23.235004000 10.7957
192.168.0.15:50986 <--> 173.194.127.198:80 6 2,931 bytes 8 1,549 bytes 14 4,480 bytes 27.890938000 31.3471

```



```
(water@kali)-[~/SBT-DF203-Lab9]
$ tshark -r "S06C" -Y 'ip' -T fields -e Frame.number -e wlan.sa -e wlan.da -e ip.src -e ip.dst -e _ws.col.Protocol | head -n 200 | tee reports/ip_mac_mapping_sample.tsv
7 0.0.0.0 255.255.255.255 DHCP
9 0.0.0.0 255.255.255.255 DHCP
11 192.168.0.1 255.255.255.255 DHCP
20 192.168.0.15 8.8.8.8 DNS
21 192.168.0.15 8.8.8.8 DNS
22 192.168.0.15 224.0.0.2 IGMPv2
23 192.168.0.15 224.0.0.251 IGMPv2
24 192.168.0.15 224.0.0.2 IGMPv2
25 192.168.0.15 224.0.0.251 IGMPv2
26 8.8.8.8 192.168.0.15 DNS
27 8.8.8.8 192.168.0.15 DNS
30 192.168.0.15 17.172.239.31 TCP
31 192.168.0.15 17.172.239.127 TCP
33 17.172.239.31 192.168.0.15 TCP
35 192.168.0.15 17.172.239.31 TCP
36 192.168.0.15 17.172.239.31 TLSv1
37 17.172.239.127 192.168.0.15 TCP
38 192.168.0.15 17.172.239.127 TCP
42 17.172.239.31 192.168.0.15 TCP
43 17.172.239.31 192.168.0.15 TLSv1
44 17.172.239.31 192.168.0.15 TLSv1
46 17.172.239.31 192.168.0.15 TLSv1
47 0.0.0.0 255.255.255.255 DHCP
48 0.0.0.0 255.255.255.255 DHCP
49 192.168.0.15 17.172.239.31 TCP
50 192.168.0.15 17.172.239.31 TCP
51 192.168.0.1 192.168.0.15 DHCP
52 192.168.0.15 17.172.239.31 TLSv1
55 17.172.239.31 192.168.0.15 TCP
56 192.168.0.15 17.172.239.31 TLSv1
57 17.172.239.31 192.168.0.15 TCP
58 17.172.239.31 192.168.0.15 TLSv1
59 192.168.0.15 17.172.239.31 TCP
60 192.168.0.15 17.172.239.31 TLSv1
61 17.172.239.31 192.168.0.15 TLSv1
65 192.168.0.15 17.172.239.31 TCP
66 192.168.0.15 17.172.239.31 TLSv1
69 17.172.239.31 192.168.0.15 TCP
70 192.168.0.15 17.172.239.31 TLSv1
74 17.172.239.31 192.168.0.15 TCP
75 17.172.239.31 192.168.0.15 TLSv1
76 192.168.0.15 17.172.239.31 TCP
81 192.168.0.15 224.0.0.251 MDNS
83 192.168.0.15 224.0.0.251 MDNS
85 192.168.0.15 224.0.0.251 MDNS
87 192.168.0.15 224.0.0.251 MDNS
91 192.168.0.15 224.0.0.251 MDNS
93 192.168.0.15 224.0.0.251 MDNS
```

```
102 192.168.0.15 198.41.209.137 HTTP
103 198.41.209.137 192.168.0.15 TCP
104 192.168.0.15 224.0.0.251 MDNS
106 192.168.0.15 224.0.0.251 MDNS
107 198.41.209.137 192.168.0.15 TCP
108 198.41.209.137 192.168.0.15 TCP
109 198.41.209.137 192.168.0.15 TCP
110 198.41.209.137 192.168.0.15 TCP
111 198.41.209.137 192.168.0.15 TCP
112 198.41.209.137 192.168.0.15 TCP
113 198.41.209.137 192.168.0.15 TCP
114 198.41.209.137 192.168.0.15 TCP
115 198.41.209.137 192.168.0.15 TCP
116 198.41.209.137 192.168.0.15 HTTP
117 192.168.0.15 198.41.209.137 TCP
118 192.168.0.15 198.41.209.137 TCP
119 192.168.0.15 198.41.209.137 TCP
120 192.168.0.15 198.41.209.137 TCP
121 192.168.0.15 198.41.209.137 TCP
122 192.168.0.15 198.41.209.137 TCP
123 192.168.0.15 198.41.209.137 TCP
124 192.168.0.15 198.41.209.137 TCP
125 192.168.0.15 198.41.209.137 TCP
126 192.168.0.15 198.41.209.137 TCP
127 192.168.0.15 8.8.8.8 DNS
128 8.8.8.8 192.168.0.15 DNS
129 192.168.0.15 23.21.196.154 TCP
130 23.21.196.154 192.168.0.15 TCP
131 192.168.0.15 23.21.196.154 TCP
132 192.168.0.15 23.21.196.154 HTTP
133 192.168.0.15 23.21.196.154 TCP
134 17.172.232.98 192.168.0.15 TLSv1
135 192.168.0.15 17.172.232.98 TCP
136 192.168.0.15 8.8.8.8 DNS
137 8.8.8.8 192.168.0.15 DNS
138 192.168.0.15 199.27.79.193 TCP
139 192.168.0.15 199.27.79.193 TCP
140 192.168.0.15 199.27.79.193 TCP
141 192.168.0.15 199.27.79.193 HTTP
142 199.27.79.193 192.168.0.15 TCP
143 199.27.79.193 192.168.0.15 TCP
144 199.27.79.193 192.168.0.15 TCP
145 199.27.79.193 192.168.0.15 TCP
146 199.27.79.193 192.168.0.15 TCP
147 199.27.79.193 192.168.0.15 TCP
148 199.27.79.193 192.168.0.15 TCP
149 199.27.79.193 192.168.0.15 TCP
150 199.27.79.193 192.168.0.15 TCP
151 199.27.79.193 192.168.0.15 TCP
152 199.27.79.193 192.168.0.15 TCP
153 192.168.0.15 199.27.79.193 TCP
154 192.168.0.15 199.27.79.193 TCP
155 192.168.0.15 199.27.79.193 TCP
156 192.168.0.15 199.27.79.193 TCP
157 192.168.0.15 199.27.79.193 TCP
```

```
(water@kali)-[~/SBT-DF203-Lab9]
$ ls -lh reports/endpoint* reports/tcp_conversations.txt reports/ip_mac_mapping_sample.tsv reports/decrypted_capture_path.txt
-rw-rw-r-- 1 water water 44 Sep 25 17:02 reports/decrypted_capture_path.txt
-rw-rw-r-- 1 water water 2.0K Sep 25 17:03 reports/ethernet_endpoints.txt
-rw-rw-r-- 1 water water 5.9K Sep 25 17:03 reports/ip_endpoints.txt
-rw-rw-r-- 1 water water 7.2K Sep 25 17:07 reports/ip_mac_mapping_sample.tsv
-rw-rw-r-- 1 water water 13K Sep 25 17:06 reports/tcp_conversations.txt

(water@kali)-[~/SBT-DF203-Lab9]
```

13. Part F - Extract Images, HTML and Other Objects

5. First try Wireshark File > Export Objects for HTTP or other recognized protocols.
6. Use tshark object export for HTTP where available.

7. Use foremost as a carving method and document that carved files may include false positives or lack original filenames.

8. Open recovered objects only inside the lab VM and hash them before viewing.

```
mkdir -p exported/http_objects exported/foremost
```

```
# HTTP object export if HTTP is present
```

```
tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee
```

```
reports/http_export_log.txt
```

```
# Generic carving
```

```
foremost -i "$DEC" -o exported/foremost | tee reports/foremost_log.txt
```

```
find exported -type f -exec file {} \; | tee reports/exported_file_types.txt
```

```
find exported -type f -exec sha256sum {} \; | tee reports/exported_file_hashes.txt
```

```
(water@kali)~[~/SBT-DF203-Lab9]
$ echo $(cat working/file_working-dec)
Decrypted capture: working/file_working-dec
$ ls -lh working
total 39M
-rw-rw-r-- 1 water water 14M Sep 25 10:17 file_working
-rw-rw-r-- 1 water water 13M Sep 25 16:50 file_working-dec
-rw-rw-r-- 1 water water 13M Sep 25 10:17 file_working.xz
(water@kali)~[~/SBT-DF203-Lab9]
$ mkdir -p exported/http_objects exported/foremost
(water@kali)~[~/SBT-DF203-Lab9]
$ ls -ld exported/http_objects exported/foremost
drwxrwxr-x 4 water water 4096 Sep 25 17:14 exported
drwxrwxr-x 2 water water 4096 Sep 25 17:14 exported/foremost
drwxrwxr-x 2 water water 4096 Sep 25 17:14 exported/http_objects
(water@kali)~[~/SBT-DF203-Lab9]
$
```

```
(water@kali)~[~/SBT-DF203-Lab9]
$ tshark -r "$DEC" --export-objects http,exported/http_objects 2>&1 | tee reports/http_export_log.txt
1 0.000000 EFMNetworks:55:97:d4 → Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.1
2 0.943674 Apple:68:96:7c → NetisTechnol:48:e5:b4 ARP 42 Who has 192.168.1.1? Tell 192.168.1.41
3 0.944186 Apple:68:96:7c → NetisTechnol:48:a8:b5 ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
4 0.944186 Apple:68:96:7c → 08:10:77:92:7c:2f ARP 42 Who has 192.168.1.1? Tell 192.168.1.12
5 0.944186 Apple:68:96:7c → EFMNetworks:55:97:d4 ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
6 0.944186 Apple:68:96:7c → TendaTechnol:58:c4:ba ARP 42 Who has 192.168.0.1? Tell 192.168.10.113
7 0.944185 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e6be1c
8 0.944704 EFMNetworks:55:97:d4 → Apple:68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97:d4
9 0.947778 0.0.0.0 → 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e6be1c
10 0.949306 fe80::1c0c:f6eb:8fb8e → ff02::2 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
11 0.954435 192.168.0.1 → 255.255.255.255 DHCP 590 DHCP NAK - Transaction ID 0x21e6be1c
12 0.954937 Apple:68:96:7c → Broadcast ARP 42 ARP Announcement for 192.168.0.15
13 0.955970 fe80::1c0c:f6eb:8fb8e → ff02::2 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
14 0.956994 Apple:68:96:7c → Broadcast ARP 42 ARP Announcement for 192.168.0.15
15 0.959034 Apple:68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
16 0.960066 Apple:68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
17 1.026106 Apple:68:96:7c → Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
18 1.026909 Apple:68:96:7c → Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
19 1.026996 EFMNetworks:55:97:d4 → Apple:68:96:7c ARP 42 192.168.0.1 is at 00:26:66:55:97:d4
20 1.030202 192.168.0.15 → 8.8.8.8 DNS 73 Standard query 0x9583 A www.apple.com
21 1.186368 192.168.0.15 → 8.8.8.8 DNS 85 Standard query 0xbe40 A 48-courier.push.apple.com
22 1.187392 192.168.0.15 → 224.0.0.2 IGMPv2 46 Leave Group 224.0.0.251
23 1.187392 192.168.0.15 → 224.0.0.251 IGMPv2 46 Membership Report group 224.0.0.251
24 1.187904 192.168.0.15 → 224.0.0.2 IGMPv2 46 Leave Group 224.0.0.251
25 1.189439 192.168.0.15 → 224.0.0.251 IGMPv2 46 Membership Report group 224.0.0.251
26 1.350208 8.8.8.8 → 192.168.0.15 DNS 203 Standard query response 0x9583 A www.apple.com CNAME www.isg-apple.com.akadns.net CNAME www.apple.com.edgekey.net CNAME e3191.dscc.akamaiedge.net A 10
4.74.154.228
27 1.350718 8.8.8.8 → 192.168.0.15 DNS 299 Standard query response 0xbe40 A 48-courier.push.apple.com CNAME 48.courier-push-apple.com.akadns.net CNAME us-courier.push-apple.com.akadns.net A 17.
172.239.31 A 17.172.239.127 A 17.172.232.188 A 17.172.239.83 A 17.172.232.166 A 17.172.239.25 A 17.172.232.170 A 17.172.232.185
28 1.380720 Apple:68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
29 1.351744 Apple:68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
30 1.353792 192.168.0.15 → 17.172.239.31 TCP 78 50954 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=16 TSval=549602880 TSecr=0 SACK_PERM
31 1.556032 192.168.0.15 → 17.172.239.127 TCP 78 50955 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=16 TSval=549603081 TSecr=0 SACK_PERM
32 1.703489 Apple:68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
33 1.703487 17.172.239.31 → 192.168.0.15 TCP 74 443 → 50954 [SYN, ACK] Seq=0 Ack=1 Win=14480 Len=0 MSS=1460 SACK_PERM TSval=646109317 TSecr=549602880 WS=128
34 1.704513 Apple:68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
35 1.729889 192.168.0.15 → 17.172.239.31 TCP 66 50954 → 443 [ACK] Seq=1 Ack=1 Win=131760 Len=0 TSval=549603254 TSecr=646109317
36 1.743425 192.168.0.15 → 17.172.239.31 TLSv1 255 client Hello (SNI=courier.push.apple.com)
37 1.79168 17.172.239.127 → 192.168.0.15 TCP 74 443 → 50955 [SYN, ACK] Seq=0 Ack=1 Win=14480 Len=0 MSS=1460 SACK_PERM TSval=641873548 TSecr=549603081 WS=128
38 1.906816 192.168.0.15 → 17.172.239.127 TCP 54 50955 → 443 [RST] Seq=1 Win=0 Len=0
39 1.923199 Apple:68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
40 1.924265 Apple:68:96:7c → Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
41 2.054781 fe80::1c0c:f6eb:8fb8e → ff02::16 ICMPv6 90 Multicast Listener Report Message v2
42 2.054784 17.172.239.31 → 192.168.0.15 TCP 66 443 → 50954 [ACK] Seq=1 Ack=190 Win=15616 Len=0 TSval=646109707 TSecr=549603268
43 2.054784 17.172.239.31 → 192.168.0.15 TLSv1 1514 Server Hello
44 2.054785 17.172.239.31 → 192.168.0.15 TLSv1 1514 Certificate
45 2.055297 17.172.239.31 → 192.168.0.15 TLSv1 194 Server Key Exchange, Certificate Request, Server Hello Done
46 2.056834 fe80::1c0c:f6eb:8fb8e → ff02::16 ICMPv6 90 Multicast Listener Report Message v2
```



```
46 2.056834 fe80::1c0c:f6e8:fedb:8f8e > ff02::16 ICMPv6 90 Multicast Listener Report Message v2
47 2.088572 0.0.0.0 > 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e0be1d
48 2.092161 0.0.0.0 > 255.255.255.255 DHCP 342 DHCP Request - Transaction ID 0x21e0be1d
49 2.094716 192.168.0.15 > 17.172.239.31 TCP 66 50954 > 443 [ACK] Seq=198 Ack=2897 Win=128864 Len=0 TSval=549603611 TSecr=646109715
50 2.094716 192.168.0.15 > 17.172.239.31 TCP 66 50954 > 443 [ACK] Seq=198 Ack=3025 Win=128736 Len=0 TSval=549603611 TSecr=646109715
51 2.095232 192.168.0.15 > 192.168.0.15 TLSv1 125 Change Cipher Spec, Encrypted Handshake Message
52 2.229373 192.168.0.15 > 17.172.239.31 TLSv1 840 Certificate
53 2.255997 Apple.68:96:7c > Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
54 2.257026 Apple.68:96:7c > Broadcast ARP 42 Who has 169.254.255.255? Tell 192.168.0.15
55 2.580006 17.172.239.31 > 192.168.0.15 TCP 66 443 > 50954 [ACK] Seq=3025 Ack=964 Win=17152 Len=0 TSval=646110233 TSecr=549603649
56 2.608253 192.168.0.15 > 17.172.239.31 TLSv1 339 Client Key Exchange, Certificate Verify, Change Cipher Spec, Encrypted Handshake Message
57 2.930878 17.172.239.31 > 192.168.0.15 TCP 66 443 > 50954 [ACK] Seq=3025 Ack=1237 Win=18688 Len=0 TSval=646110571 TSecr=549604128
58 2.930878 17.172.239.31 > 192.168.0.15 TLSv1 125 Change Cipher Spec, Encrypted Handshake Message
59 2.960064 192.168.0.15 > 17.172.239.31 TCP 66 50954 > 443 [ACK] Seq=1237 Ack=3084 Win=131008 Len=0 TSval=549604479 TSecr=646110573
60 2.961088 192.168.0.15 > 17.172.239.31 TLSv1 199 Application Data
61 3.283645 17.172.239.31 > 192.168.0.15 TLSv1 119 Application Data
62 3.311294 Apple.68:96:7c > Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
63 3.312320 Apple.68:96:7c > Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
64 3.312318 EFMNetworks_55:97:d4 > Apple.68:96:7c ARP 42 192.168.0.1 is at 00:26:06:55:97:d4
65 3.313343 192.168.0.15 > 17.172.239.31 TCP 66 50954 > 443 [ACK] Seq=1370 Ack=3137 Win=131008 Len=0 TSval=549604829 TSecr=646110924
66 3.314878 192.168.0.15 > 17.172.239.31 TLSv1 103 Application Data
67 3.459264 Apple.68:96:7c > Broadcast ARP 42 ARP Announcement for 192.168.0.15
68 3.460287 Apple.68:96:7c > Broadcast ARP 42 ARP Announcement for 192.168.0.15
69 3.635388 17.172.239.31 > 192.168.0.15 TCP 66 443 > 50954 [ACK] Seq=3137 Ack=1407 Win=20224 Len=0 TSval=646111318 TSecr=549604833
70 3.663552 192.168.0.15 > 17.172.239.31 TLSv1 321 Application Data, Application Data
71 3.810560 Apple.68:96:7c > Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
72 3.811581 Apple.68:96:7c > Broadcast ARP 42 Who has 192.168.0.1? Tell 192.168.0.15
73 3.811579 EFMNetworks_55:97:d4 > Apple.68:96:7c ARP 42 192.168.0.1 is at 00:26:06:55:97:d4
74 3.987707 17.172.239.31 > 192.168.0.15 TCP 66 443 > 50954 [ACK] Seq=3137 Ack=1662 Win=21760 Len=0 TSval=646111627 TSecr=549605181
75 3.987707 17.172.239.31 > 192.168.0.15 TLSv1 103 Application Data
76 4.015358 fe80::1c0c:f6e8:fedb:8f8e > ff02::16 ICMPv6 90 Multicast Listener Report Message v2
77 4.016892 fe80::1c0c:f6e8:fedb:8f8e > ff02::16 ICMPv6 90 Multicast Listener Report Message v2
78 4.016895 fe80::1c0c:f6e8:fedb:8f8e > ff02::16 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
79 5.743932 fe80::1c0c:f6e8:fedb:8f8e > ff02::12 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
80 5.744955 fe80::1c0c:f6e8:fedb:8f8e > ff02::12 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
81 7.023344 192.168.0.15 > 224.0.0.251 MDNS 146 Standard query response 0x0000 TXT OPT
82 7.025592 fe80::1c0c:f6e8:fedb:8f8e > ff02::fb MDNS 166 Standard query response 0x0000 TXT OPT
83 7.027133 192.168.0.15 > 224.0.0.251 MDNS 146 Standard query response 0x0000 TXT OPT
84 7.029188 fe80::1c0c:f6e8:fedb:8f8e > ff02::fb MDNS 166 Standard query response 0x0000 TXT OPT
85 8.023553 fe80::1c0c:f6e8:fedb:8f8e > ff02::fb MDNS 146 Standard query response 0x0000 TXT OPT
86 8.023553 fe80::1c0c:f6e8:fedb:8f8e > ff02::fb MDNS 166 Standard query response 0x0000 TXT OPT
87 8.025086 192.168.0.15 > 224.0.0.251 MDNS 146 Standard query response 0x0000 TXT OPT
88 8.027134 fe80::1c0c:f6e8:fedb:8f8e > ff02::fb MDNS 166 Standard query response 0x0000 TXT OPT
89 8.809050 EFMNetworks_55:97:d4 > Apple.68:96:7c ARP 42 Who has 192.168.0.15? Tell 192.168.0.15
90 8.870983 Apple.68:96:7c > EFMNetworks_55:97:d4 ARP 42 192.168.0.15 is at f0:f6:1c:68:96:7c
91 9.997440 192.168.0.15 > 224.0.0.251 MDNS 146 Standard query response 0x0000 TXT OPT
92 9.997440 fe80::1c0c:f6e8:fedb:8f8e > ff02::fb MDNS 166 Standard query response 0x0000 TXT OPT
93 10.005121 192.168.0.15 > 224.0.0.251 MDNS 146 Standard query response 0x0000 TXT OPT
94 10.007680 fe80::1c0c:f6e8:fedb:8f8e > ff02::fb MDNS 166 Standard query response 0x0000 TXT OPT
95 10.303616 fe80::1c0c:f6e8:fedb:8f8e > ff02::12 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
96 10.304640 fe80::1c0c:f6e8:fedb:8f8e > ff02::12 ICMPv6 70 Router Solicitation from f0:f6:1c:68:96:7c
97 13.465404 192.168.0.15 > 8.8.8.8 DNS 72 Standard query 0xf04d A m.reddit.com
98 13.641024 8.8.8.8 > 192.168.0.15 DNS 312 Standard query response 0xf04d A m.reddit.com A 198.41.209.137 A 198.41.208.141 A 198.41.209.136 A 198.41.209.142 A
198.41.208.143 A 198.41.208.139 A 198.41.209.140 A 198.41.209.139 A 198.41.209.141 A 198.41.209.143 A 198.41.208.138 A 198.41.208.142
99 13.669693 192.168.0.15 > 198.41.209.137 TCP 78 50956 > 80 [SWN] Seq=0 Win=65535 Len=0 MSS=1460 WS=16 TSval=549615185 TSecr=0 SACK PERM
```

```
(water@kali) [~/SBT-DF203-Lab9]
$ find exported/http_objects -type f -ls
1615279 4 -rw-r--r-- 1 water water 35 Sep 25 17:16 exported/http_objects/6(1).gif
1615273 4 -rw-r--r-- 1 water water 216 Sep 25 17:16 exported/http_objects/redir.html
1615266 4 -rw-r--r-- 1 water water 4 Sep 25 17:16 exported/http_objects/figure-10.png
countIdk2k3AnullC2C22sessionIdk22k3A224d0d20Seea8f1f1bdsb098028b761dc22k2C22p2atform2k23A22web2k23C22source2k23A2X(3).com2F8iVyt6x22X7D
1615289 76 -rw-r--r-- 1 water water 74536 Sep 25 17:16 exported/http_objects/X2f(1)
1615305 4 -rw-r--r-- 1 water water 132 Sep 25 17:16 exported/http_objects/superfish.args.min.js
1615320 156 -rw-r--r-- 1 water water 15962 Sep 25 17:16 exported/http_objects/figure-10.png
1615268 4 -rw-r--r-- 1 water water 4 Sep 25 17:16 exported/http_objects/lumbar.gifX3f8a-X7B822UID22k3A222ae1543cd8ab60f50e1849b3db2c79e9ce5872c0878dfc88d838afac1a3d32k2C22acc
countIdk2k3AnullC2C22sessionIdk22k3A224d0d20Seea8f1f1bdsb098028b761dc22k2C22p2atform2k23A22web2k23C22source2k23A2X(4).com2F8iVyt6x22X7D
1615304 20 -rw-r--r-- 1 water water 19977 Sep 25 17:16 exported/http_objects/portfolio-slideshow.min.js
1615291 4 -rw-r--r-- 1 water water 35 Sep 25 17:16 exported/http_objects/collect33Fv16_vw338a420489676t-pageview6_s=16d1httpX3A22FX2Fimgur.com2F8iVyt66dr=htpX3A22FX2Fm.reddit
.com2F8ul=usde=UTF-8&dt=Epic220doggileX20-X20Imgurds-32-bit6sr-320x568vvp-980x14966je=06_utm=2473412.14055857036tId=UA-6671088-26z=44004997
1615189 4 -rw-r--r-- 1 water water 210 Sep 25 17:16 exported/http_objects/hit.json
1615322 8 -rw-r--r-- 1 water water 5361 Sep 25 17:16 exported/http_objects/og15PA(1).mp4
1615293 4 -rw-r--r-- 1 water water 247 Sep 25 17:16 exported/http_objects/css33family=Oswald&7C
1615232 4 -rw-r--r-- 1 water water 8 Sep 25 17:16 exported/http_objects/bidX3fsrc=30796w=htpX3A22FX2Fimgur.com2F8iVyt66cb=3908366
1615267 4 -rw-r--r-- 1 water water 141 Sep 25 17:16 exported/http_objects/v2(2)
1615314 4 -rw-r--r-- 1 water water 110 Sep 25 17:16 exported/http_objects/ui-b6_glass_75_e6e6e_1x400.png
1615297 4 -rw-r--r-- 1 water water 1018 Sep 25 17:16 exported/http_objects/upw-theme-standard.min.css
1615309 16 -rw-r--r-- 1 water water 13224 Sep 25 17:16 exported/http_objects/Zd2E9abXLFgs9G3YK2MxNB08B5xanTJV5B9YUJQ.woff
1615269 4 -rw-r--r-- 1 water water 302 Sep 25 17:16 exported/http_objects/6au1d=372082708res=320x568x326p1g=qtX2Cpm6ch=UTF-8&tze=5400ws=300x250
1615287 4 -rw-r--r-- 1 water water 2 Sep 25 17:16 exported/http_objects/og15PA(1).mp4
1615311 24 -rw-r--r-- 1 water water 28224 Sep 25 17:16 exported/http_objects/YDaaLskQ05MD0AgvHUQCclQa6gm6b580u2Qn-1Plo1Go.woff
1615323 4 -rw-r--r-- 1 water water 685 Sep 25 17:16 exported/http_objects/Analytics3fver=36visitor_id=6pi_opt_in=6campaign_id=15646account_id=473826title=Decoding%20Zeus%20Disguised%20as%20an%20RtfX20F1eX20-X20PhishMeurl=htpX3A22FX2Fphishme.com2F8decoding=zeus-disguised=an-rtf-fileX20foreferrer=htpX3A22.com2F8F2fnetsec
%20disguised%20as%20an%20RtfX20F1eX20-X20PhishMeurl=htpX3A22FX2Fphishme.com2F8decoding=zeus-disguised=an-rtf-fileX20foreferrer=htpX3A22.com2F8F2fnetsec
1615295 4 -rw-r--r-- 1 water water 1938 Sep 25 17:16 exported/http_objects/trackalyze.js
1615319 216 -rw-r--r-- 1 water water 219223 Sep 25 17:16 exported/http_objects/figure-5.png
1615286 0 -rw-r--r-- 1 water water 0 Sep 25 17:16 exported/http_objects/cs.js
1610780 76 -rw-r--r-- 1 water water 74536 Sep 25 17:16 exported/http_objects/X2f
1615303 12 -rw-r--r-- 1 water water 12076 Sep 25 17:16 exported/http_objects/tabs.min.js
1615296 8 -rw-r--r-- 1 water water 7327 Sep 25 17:16 exported/http_objects/portfolio-slideshow.min.css
1615326 104 -rw-r--r-- 1 water water 104015 Sep 25 17:16 exported/http_objects/bootsrap.min.css
1615282 8 -rw-r--r-- 1 water water 4632 Sep 25 17:16 exported/http_objects/imageViewerinline.jsX3f1425580383
1615276 4 -rw-r--r-- 1 water water 3799 Sep 25 17:16 exported/http_objects/iframe.html
1615313 40 -rw-r--r-- 1 water water 40278 Sep 25 17:16 exported/http_objects/dc.js
1615299 4 -rw-r--r-- 1 water water 757 Sep 25 17:16 exported/http_objects/comment-reply.min.js
1615284 84 -rw-r--r-- 1 water water 82382 Sep 25 17:16 exported/http_objects/og15PA.jpg
1615291 4 -rw-r--r-- 1 water water 67 Sep 25 17:16 exported/http_objects/of_destiny.pngX3fV=h2Bw3qHw8Nrlcs96j8ZKx2BqrueW91KTg3Y9jHy9f3jgs2432FhnEnoQvPwm1221H67UGAAV5EK3D
1610789 4 -rw-r--r-- 1 water water 35 Sep 25 17:16 exported/http_objects/analytic33fconly=true&visitor_id=736851146pl_opt_in=6campaign_id=15646account_id=473826title=Decoding%20Zeus%20Disguised%20as%20an%20RtfX20F1eX20-X20PhishMeurl=htpX3A22FX2Fphishme.com2F8decoding=zeus-disguised=an-rtf-fileX20foreferrer=htpX3A22.com2F8F2fnetsec
enc=ndst=0et=1425628475388;tz=+540;ref=htpX3A22FX2Fm.reddit.com2F8url=htpX3A22FX2Fimgur.com2F8iVyt66;ogl=url.htt.photocredX20x40theBarkhaus
1615290 68 -rw-r--r-- 1 water water 66008 Sep 25 17:16 exported/http_objects/netsec
1615329 4 -rw-r--r-- 1 water water 3573 Sep 25 17:16 exported/http_objects/shal.js
1615310 16 -rw-r--r-- 1 water water 12760 Sep 25 17:16 exported/http_objects/qpy-iU1kPw-Vf0d6cs6re-vvDin1pKkatttpeZ5c0A.woff
1615301 6 -rw-r--r-- 1 water water 2204 Sep 25 17:16 exported/http_objects/css33family=Noto&ans=33AA0052C4001talic2C7005CYanone&Kaffeeatz
%3A400%2C7005ver=3.8.1
1615294 72 -rw-r--r-- 1 water water 71547 Sep 25 17:16 exported/http_objects/dynamik-min.css
1615244 4 -rw-r--r-- 1 water water 4 Sep 25 17:16 exported/http_objects/lumbar.gifX3f8a-X7B822UID22k3A222ae1543cd8ab60f50e1849b3db2c79e9ce5872c0878dfc88d838afac1a3d32k2C22acc
countIdk2k3AnullC2C22sessionIdk22k3A224d0d20Seea8f1f1bdsb098028b761dc22k2C22p2atform2k23A22web2k23C22source2k23A2X(1).com2F8iVyt6x22X7D
```

```
(water@kali) [~/SBT-DF203-Lab9]
$ foremost -i 'SMB*' -o exported/foremost | tee reports/Foremost_log.txt
Processing: working/file_working-dec
[+]

(water@kali) [~/SBT-DF203-Lab9]
$ find exported/foremost -type f -ls
1615374 88 -rw-r--r-- 1 water water 89458 Sep 25 17:22 exported/foremost/jpg/00004538.jpg
1615373 260 -rw-r--r-- 1 water water 264865 Sep 25 17:22 exported/foremost/jpg/0000104.jpg
1615372 4 -rw-r--r-- 1 water water 1606 Sep 25 17:22 exported/foremost/audit.txt
1615381 4 -rw-r--r-- 1 water water 1322 Sep 25 17:22 exported/foremost/htm/00024227.htm
1615382 4 -rw-r--r-- 1 water water 67 Sep 25 17:22 exported/foremost/htm/0001639.htm
1615375 5 -rw-r--r-- 1 water water 35 Sep 25 17:22 exported/foremost/gif/00000406.gif
1615381 4 -rw-r--r-- 1 water water 35 Sep 25 17:22 exported/foremost/gif/00010876.gif
1615376 4 -rw-r--r-- 1 water water 35 Sep 25 17:22 exported/foremost/gif/00001033.gif
1615377 4 -rw-r--r-- 1 water water 42 Sep 25 17:22 exported/foremost/gif/00002450.gif
1615379 4 -rw-r--r-- 1 water water 35 Sep 25 17:22 exported/foremost/gif/00003719.gif
1615380 4 -rw-r--r-- 1 water water 49 Sep 25 17:22 exported/foremost/gif/00010456.gif
1615376 4 -rw-r--r-- 1 water water 35 Sep 25 17:22 exported/foremost/gif/00010260.gif
1615387 4 -rw-r--r-- 1 water water 110 Sep 25 17:22 exported/foremost/png/00010071.png
1615384 4 -rw-r--r-- 1 water water 283 Sep 25 17:22 exported/foremost/png/00001765.png
1615385 4 -rw-r--r-- 1 water water 166 Sep 25 17:22 exported/foremost/png/00001831.png
1615386 4 -rw-r--r-- 1 water water 67 Sep 25 17:22 exported/foremost/png/00010214.png
1615389 4 -rw-r--r-- 1 water water 598 Sep 25 17:22 exported/foremost/png/00001242.png
1615388 4 -rw-r--r-- 1 water water 425 Sep 25 17:22 exported/foremost/png/00020736.png
```


[illegible]


```
65330bc2d3d72817583024132b7f5cfd35eef0252d6ec1b9c9a4e13bb3 exported/http_objects/v2
075457fca80e12d43f9c8b7967692a28f0c7cabbf15e10037a148729d exported/http_objects/hit(1).json
02e357aa1b284a9e6a710885744310191740a96d99bb18916b1cbc547ea79e exported/http_objects/x2f(2)
59ebc47331c997b569e83c98089d4b1f379e2bec0e6cf979f8c7e43643f452 exported/http_objects/page-list.css
58a6aea4154172376ee4b902ef7db68cba24dac17b679da899e460ef8873 exported/http_objects/sharePlayer.css%3f1425580383
96a296d2247285c67be49c3d38f8a517f76a35dc5b87e410b78630a89c7c exported/http_objects/og15SPA.mp4
3590f1e1f0080fe42d63eeb14f476884dc5b7d7c0ef3c3f69227086f5bd73a exported/http_objects/jquery-ui.css
d45834052695f8c21036c7278a47b0bca00ae404982e554967103381477 exported/http_objects/v2(1)
6e5e202f1f8c9b5b489f11d21d94c064b9d70c1ae100f1c7248dc121a85c07d exported/http_objects/superfish.min.js
ea2e29625de346345e93b0b2b065f5833e05b977f7a052b1c141e754d62e1a8b exported/http_objects/ui-icons_888888_256x240.png
dd72b0b714d6ca5c346b9f8b11f88a1c16135723b08a04c2b0863db3a30 exported/http_objects/Adx3AdxGdp_req-16Correlator-3971610776330246output=json_htmlcallback=callbackProxy6impl=fifs6json_a=16eid=1088090
3585fV-1-0-26iu_parts=476166082CH_I_300x2506enc_prev_ius=x2F0x2F1b exported/http_objects/sharePlayer.js%3f1425580383
7a69c1c1a56281b6ae35d95f1e9bba3d687e18ccf5d6759b964e232c3c80 exported/http_objects/figure-4.png
9f2ee139f30b4d475746201ee5c047c45ff613b5f9b8f5b5e1474f86980cb exported/http_objects/lumbar.gif%3fa=x7B322UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c087dffc88d838afa4c1a3d%22%22accountId
74234e98afe7498fb5daf1f36ac2d78ac339464f950703b8c019892f982b90b exported/http_objects/lumbar.gif%3fa=x7B322UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c087dffc88d838afa4c1a3d%22%22accountId
k223Anu1l32C22sessionid%22%3A%224d0d205eeaf1f1b5b098028b761dc22%3A%22platform%22%3A%22web%22%3A%22source%22%3A%22f.com%2F81vy7t6x22%7D exported/http_objects/adc-i-00aecd.png
534c09b9b2d436a38c26aeadd2679d2a2c97d4962399a8da227ee07162a755 exported/http_objects/adc-i-00aecd.png
ef1955ae757c8b966c8324835031bd3a30f658ced1f1387f8ebf05ab3368629 exported/http_objects/activeview%3fid=osdmbavi=BLAOYQV35VNmMHTy9QwknIgcCADA_Sou=GEAABAOHAIAQLgAgDIASke4A0BA0CwMDEIAB6ti=16adk=109919
5366p=784,685,1034,9850tos=1163,0,0,0,0,0mtos=1163,1163,116,com%2Fads%3Fpage%3Dinside%26main%3D1326hot%3D%26time%3D142560863661s=300,2506uc=5 exported/http_objects/hl583fversion=4.12
0808db07799acd281fcb3b46c9708dae1f83b8e91f9551185394abf19d4b797ae exported/http_objects/hl583fversion=4.12
49a6ae42b1b3af56c164ac3b0dd2f8fcedd2f13a5c08c25521606ab33ae5 exported/http_objects/b90BGL01M2705pmcxE8mPZRQUBYfRc-1FWT9T9EE8M.woff
8337212354871836e6763a41e015916c89bac5b3f1f0adf60ba43c7c806e1015 exported/http_objects/_utm.gif%3futmmv=5.6.3dcutms=10utmn=1196617678utmh=phishme.com6utms=UTF-88utms=320x568utmp=320x460utmsc=32
-bitutml=en-usbutmje=86utmlf=-butmdf=Decoding%20ZeUS%20.com%7Cutmcn%3D(referral)%7Cutmcnd%3Dreferral%7Cutmcct%3D%2F%32Fnetsec%3B8utmu=q-
8337212354871836e6763a41e015916c89bac5b3f1f0adf60ba43c7c806e1015 exported/http_objects/6.gif
354065559b78314d2d18d90751916434bf8c20a18565cde6747a7a1d59 exported/http_objects/blah.css
8337e56783c41746a9f3612637509560424617e3135248640eac142d6e0194 exported/http_objects/dot.gif
4cde20607a09eef64289f5775cbaca271bd09ca6f9bfbaf615876ee6f9257 exported/http_objects/pdx3fcc=16plm=56ph=d14b94c9-e278-4d1d-87a0-a6a729350974
74234e98afe7498fb5daf1f36ac2d78ac339464f950703b8c019892f982b90b exported/http_objects/imageview.gif%3fa=ogl15PA8r=http%3A%2FX2Fm.reddit.com%2F
055154927bba6b4a204c7142f3945a40806cb9b526e76e59f8dafebc9405b exported/http_objects/accordion.min.js
77050c4edfe07cc4d83bd5d7513b07404f80663691e956c8a8c7f7eaf exported/http_objects/figure-15.png
47218e7b0736e24b7b6c71d9a46c2379c2f396c3317d7ecedbf0b99ab2c9 exported/http_objects/jquery.cookie.js
74234e98afe7498fb5daf1f36ac2d78ac339464f950703b8c019892f982b90b exported/http_objects/lumbar.gif%3fa=x7B322UID%22%3A%222ae1543dcd8ab603f50e1849b3db2c79e9ce5872c087dffc88d838afa4c1a3d%22%22accountId
k223Anu1l32C22sessionid%22%3A%224d0d205eeaf1f1b5b098028b761dc22%3A%22platform%22%3A%22web%22%3A%22source%22%3A%22f.com%2F81vy7t6x22%7D exported/foremost/jpg/00004538.jpg
f8017a7ef7652a26a543d8d7b0aebcb00676fd72ec42a2a36523a6371297dc exported/foremost/jpg/00001010.jpg
3f9bed24e04c1735b498dab0f138673c20591a5b39201e71a91dd180929ba6 exported/foremost/audit.txt
0493c100669d3d11d7002a9625a2cblfeac5c87d68edf8f0bb6533b3fa79950e exported/foremost/htm/00024427.htm
02e357aa1b284a9e6a710885744310191740a96d99bb18916b1cbc547ea79e exported/foremost/htm/00001639.htm
1d11ac950ff1be55cae699a487331563f958afd8295252ca88c96c4e72b161eb exported/foremost/gif/00000466.gif
a0d3a0af7dc2b3f3d2176fc3dcdade7ab5b987c4fed17af63550dc7cc44f8 exported/foremost/gif/00010876.gif
8337212354871836e6763a41e015916c89bac5b3f1f0adf60ba43c7c806e1015 exported/foremost/gif/00003653.gif
8337212354871836e6763a41e015916c89bac5b3f1f0adf60ba43c7c806e1015 exported/foremost/gif/00002450.gif
ef1955ae757c8b966c8324835031bd3a30f658ced1f1387f8ebf05ab3368629 exported/foremost/gif/00003719.gif
8337212354871836e6763a41e015916c89bac5b3f1f0adf60ba43c7c806e1015 exported/foremost/gif/00010456.gif
53f7e56783c41746a9f3612637509560424617e3135248640eac142d6e0194 exported/foremost/gif/00001260.gif
8337212354871836e6763a41e015916c89bac5b3f1f0adf60ba43c7c806e1015 exported/foremost/png/00010671.png
dd75d4e0ef2b185e88b0af7b6e90be74a8438ac4700658e76e754c8bf550 exported/foremost/png/00001765.png
e0c179d0b533ec5f0ac1f99d84fbcd7dc7e07acbe9defcac65416b63687950d exported/foremost/png/00001831.png
534c09b9b2d436a38c26aeadd2679d2a2c97d4962399a8da227ee07162a755 exported/foremost/png/00010214.png
e0f4635a17d19d6be4db60b7164219a3220f280013c6d631122e9d2a exported/foremost/png/00024242.png
681f62c21414bf0c7cedb61d57f950042403bc89beed026e85ae6c85d7ccf exported/foremost/png/00020736.png
792477b4d20b4cf0580011cad06d792425ce4e2c5d2321b4b41e06ac09676
```

15. WEP Security Analysis

- WEP uses RC4 with a short 24-bit IV, allowing IV reuse in busy networks.
- The same keystream reuse leaks relationships between plaintexts and enables statistical key recovery.
- WEP provides weak integrity and no modern protection against replay or key-recovery attacks.
- Recommended replacement is WPA2-AES/CCMP or WPA3 with strong authentication and protected management frames where supported.
- Legacy WEP should be removed rather than strengthened with longer passphrases.

The Lab 9 exercise showed that WEP40 is not a strong way to protect a wireless network. WEP uses a short 24-bit IV with RC4 encryption, and when the IV is reused, it can make the wireless traffic easier to analyze.

The captured traffic allowed us to study the WEP security offline, and the WEP40 key A43DF6F374 was successfully recovered and validated. This shows that someone who captures enough wireless traffic may be able to recover the WEP key without knowing the original router password.

WEP also has weak protection against changing or replaying network packets. Using a longer password does not solve these problems because the weakness is in the WEP security system itself.

For better protection, wireless networks should use WPA2-AES/CCMP or WPA3 instead of WEP. Protected management frames should also be used where the wireless equipment supports them.

Overall, the Lab 9 results showed that WEP40 is no longer suitable for protecting wireless communication because captured traffic can be studied offline and the WEP key can be recovered.

16. Required Findings Worksheet

Question	Finding
Capture format and duration	Supplied historical WEP40 capture in PCAP format. The capture contained 45,169 WLAN frames. The exact capture duration was not established in the available results.
BSSID	00:26:66:55:97:D6
Primary station MAC addresses	
Protected frame count	15713
Repeated IV evidence	Repeated IVs were investigated as evidence supporting WEP40 key recovery, but the exact repeated-IV count was not recorded in the available results.
Recovered/validated WEP40 key	A4:3D:F6:F3:74 — Airdecap format: A43DF6F374.
Decrypted capture filename/hash	Expected decrypted file: working/file_working-dec or working/file_working-dec.cap. The SHA-256 hash was not established in the available results.
Host IP and MAC	30 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 192.168.0.15 17.172.239.31 TCP

Question	Finding
	31 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 192.168.0.15 17.172.239.127 TCP 33 00:26:66:55:97:d4 f0:f6:1c:68:96:7c 17.172.239.31 192.168.0.15 TCP 35 f0:f6:1c:68:96:7c 00:26:66:55:97:d4 192.168.0.15 17.172.239.31 TCP
Server IP and MAC	1257 198.90.20.111 192.168.0.15 80 51007 1209 199.27.79.193 192.168.0.15 80 50959 1159 198.90.20.111 192.168.0.15 80 51008 1145 199.27.79.193 192.168.0.15 80 51001
Key protocols observed	WLAN traffic was observed. The protocol hierarchy contained 45,169 WLAN frames, including 15,713 data frames. Exact decrypted application protocols were not established in the available results.
Recovered images/HTML	The extraction procedure was performed/planned using Wireshark/TShark object export and Foremost carving, but the available results do not establish the exact recovered files.
Optional router passphrase	Protected appendix only. The human-readable passphrase was not included in the available results.
Security conclusion	The historical capture demonstrated that the 40-bit WEP key could be recovered and used for offline decryption. The presence of repeated IVs and successful key validation demonstrates the weakness of WEP40 for protecting wireless traffic.

2.7. Analysis and Findings

The Lab showed that WEP40 has weak security because of its short 24-bit IV and RC4 encryption. The historical capture was successfully analyzed, and the WEP40 key A43DF6F374 was recovered and validated. The BSSID identified in the capture was 00:26:66:55:97. The decrypted capture allowed the wireless traffic to be examined for endpoints, conversations, and

possible recovered objects. The results showed why WEP should not be used to protect modern wireless networks.

2.8. Challenges and Solutions

One challenge was identifying the correct decrypted capture after the WEP key recovery process. Another challenge was extracting useful information such as MAC addresses, IP addresses, and conversations from the wireless traffic. These problems were solved by using TShark commands to examine the decrypted PCAP file

2.9. Conclusion

The Lab demonstrated the weaknesses of WEP40 in a controlled historical lab environment. The WEP40 key A43DF6F374 was successfully recovered and validated from the supplied capture. The exercise showed that captured WEP traffic can be analyzed offline. It also showed that WEP weaknesses cannot be fixed simply by using a longer password. Therefore, WEP is not suitable for modern wireless security.

2.10. Recommendations

Assignment like this improved student knowledge and understanding of Network trafficking

2.11. References

Class Video Guide from Founder ICDFIA on Basic Networking Skills for Digital Forensics